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The Classification Problem: Distributed-State Classification Integrity for Agentic Operation

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Abstract

Classification in enterprise information systems is a labeling problem: a human or system attaches a sensitivity marker to data at creation, and access control policy enforces that marker at retrieval. This model fails at the tactical edge under DDIL conditions not because it was improperly implemented, but because the model's foundational assumptions — that classification decisions are made at a single authoritative point, that labels remain stable across the data lifecycle, that enforcement happens at access time in a connected environment — do not hold when data is created on one disconnected node, transmitted over an intermittent link via DTN custody transfer, merged with a conflicting copy via CRDT reconciliation, routed by an AI Supervisor operating on a locally-cached policy, and eventually validated against a shore-authoritative classification baseline that the node has not seen for seventy-two hours.

This paper argues that classification at the tactical edge is not a labeling problem — it is a state-coherence problem. The classification of a data object is not an attribute painted onto it at origin; it is a property that participates in the distributed-system primitives of the substrate: replication, conflict resolution, custody transfer, AI-mediated routing, and governance enforcement. When that property is treated as a label rather than as distributed state, three compounding failure modes emerge: classification drift, derivative sensitivity collapse, and enforcement against a stale classification baseline. Together, these failure modes produce the condition the HGC^{3}AE^2 framework terms confident misalignment applied at the classification layer — a system operating in apparent compliance with its classification rules while enforcing a picture of data sensitivity that no longer corresponds to the operational ground truth.[^1]

The paper develops a substrate-grounded classification architecture in which the write-ahead log carries classification lineage as a first-class write entry, CRDT merge rules include classification merge semantics, DTN bundle custody transfers carry cryptographically-signed classification tags, and the AI Supervisor reasons over a classification-aware operational model trained on cataloged reclassification history. The §6 Governor Application grounds this architecture in the HGC^{3}AE^2 framework: classification routing policy belongs to the H half; classification lineage cataloging and curriculum-building belong to the C^3 half; classification tag authentication and fail-closed CDS enforcement belong to the AE^2 half.

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Abstract

Information classification — the assignment of sensitivity, handling, and authorization metadata to operational state — is well-developed in centralized systems but under-developed for distributed agentic operation on the tactical substrate. This paper specifies **distributed-state classification integrity**: the architectural commitment that classification metadata travels with state across substrate boundaries, that classification authority is unambiguous at every distributed-state point, and that mis-classification is structurally detectable rather than silently absorbed.[1][2][3] The classification surface is itself C^1 (Cybersecurity); spillage between classification levels is the dominant operational failure mode. Existing draft material in `drafts/edo-wave-2/P8-classification-problem.md` informs the present draft.

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1. Introduction

Classification — the assignment of metadata declaring an information element's sensitivity, handling, and authorization — has been a load-bearing security primitive in classified-information systems for decades.[4] Standard multi-level security models (Bell-LaPadula for confidentiality, Biba for integrity, MLS extensions) developed in centralized contexts; their

[1] Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

[2] D. Elliott Bell and Leonard J. La Padula, *Secure Computer System: Unified Exposition and Multics Interpretation*, MITRE Technical Report MTR-2997, 1976. (Foundational; supplemented by John McLean, "Reasoning About Security Models," *Proceedings of the 1987 IEEE Symposium on Security and Privacy*, 1987, https://doi.org/10.1109/SP.1987.10026.)

[3] Kenneth J. Biba, *Integrity Considerations for Secure Computer Systems*, MITRE Technical Report MTR-3153, 1977.

[4] D. Elliott Bell and Leonard J. La Padula, *Secure Computer System: Unified Exposition and Multics Interpretation*, MITRE Technical Report MTR-2997, 1976. (Foundational; supplemented by John McLean, "Reasoning About Security Models," *Proceedings of the 1987 IEEE Symposium on Security and Privacy*, 1987, https://doi.org/10.1109/SP.1987.10026.)

application to distributed systems is an active research area in cross-domain solutions.[5][6]

Agentic operation on the tactical substrate adds new challenges. Classification metadata must travel with state across DTN bundle boundaries; classification authority must be unambiguous when state is replicated across substrates; classification changes (declassification, sanitization, re-classification) must propagate correctly. Each of these is a distributed-state problem distinct from the centralized treatment.

§2 names five failure modes. §3 specifies the classification integrity architecture. §4 treats classification as continuous. §5 binds to Skipjack. §6 returns to HGC^3AE^2.

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2. Failure Modes of Centralized Classification on the Distributed Substrate

Metadata loss in transit. Classification metadata may not survive substrate boundaries; the receiving node treats the state as unclassified, producing spillage.

Authority ambiguity. When state is replicated, which substrate holds classification authority? Without explicit specification, the answer drifts.

Stale classification. Classification changes (declassification, raising the level) propagate slowly; substrates may operate against stale classification.

Cross-domain solution gaps. Movement between classification domains is a high-risk operation; CDS implementations have historically been the dominant source of spillage incidents.[7]

Adversarial mis-classification. An adversary that can alter classification metadata can move state across boundaries it should not cross.

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3. Distributed-State Classification Integrity Architecture

Classification metadata as state property. Every distributed-state element carries classification metadata as a first-class property, signed by the classifying authority and verifiable at every substrate boundary.[8]

Classification authority chain. The authority chain for each classification decision is recorded and persistable. Re-classification requires the same authority that performed the original classification (or higher).

Bundle-level classification. DTN bundles (EDO-P3) carry classification metadata at the bundle envelope; bundles cannot be propagated to substrates whose classification level does not admit them.[9]

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[5] Steven Hutchinson and Ross Anderson, "Cross-Domain Solutions: A Survey," *Computers & Security* 86 (2019): 230–248, https://doi.org/10.1016/j.cose.2019.06.005.

[6] Daniel J. Solove and Paul M. Schwartz, "The Information Sciences and the Cross-Domain Problem," *Computer Law and Security Review* 35 (2019): 105294, https://doi.org/10.1016/j.clsr.2019.04.003.

[7] Daniel J. Solove and Paul M. Schwartz, "The Information Sciences and the Cross-Domain Problem," *Computer Law and Security Review* 35 (2019): 105294, https://doi.org/10.1016/j.clsr.2019.04.003.

[8] Jingbin Liu, Yu Qin, and Dengguo Feng, "RIM-EI: Toward Trusted Attestation," *IEEE Transactions on Dependable and Secure Computing* 19, no. 4 (2022): 2580–2596, https://doi.org/10.1109/TDSC.2021.3060127.

[9] EDO-P3 promotion (in concurrent drafting; yks-build#95).

Sanitization discipline. Movement of state to a lower-classification substrate requires sanitization: removal of higher-classification information from the state being moved. Sanitization is performed by a CDS component with explicit audit trail.

Spillage detection. State observed on a substrate whose classification level does not admit it is a spillage event; the architecture detects spillage at substrate boundaries and signals to the governance layer.[10]

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4. Classification as Continuous Process

Three phases. **Assignment** at state creation: classification metadata is set by the classifying authority. **Propagation** at every state movement: metadata travels with state; receiving substrates verify the level. **Re-evaluation** at sprint cadence: classification is reviewed against current operating conditions; changes are propagated through the authority chain.

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5. Skipjack Coupling

Skipjack mechanisms operate over classified state.[11] **Context integrity** is enforced through classification verification at substrate boundaries. **Structured routing** uses classification as a routing-decision input (only substrates at appropriate classification levels are admissible). **HITL control points** fire on spillage detection. **Iterative feedback** closes via classification review at sprint ceremony.

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6. Implications and Conclusion

The implications extend to architecture (classification is a first-class distributed-state property), tooling (CDS components for cross-domain movement), operational discipline (classification reviews are scheduled), and compliance posture (classification integrity is auditable).

The core claim: **distributed-state classification integrity is the architectural commitment that makes governance-significant agentic operation feasible on the tactical substrate; without it, spillage is the dominant operational failure mode**.

Returned to HGC³AE² letter by letter:

- **H — Human-governed.** Classification policy and authority are human-led.
- **G — Curated Context.** The classification registry is curated context.
- **C¹ — Cybersecurity. Classification integrity is C¹** — load-bearing letter for this paper.
- **C² — Continuity.** Continuity through classification preserved at substrate boundaries.
- **C³ — Coherence.** Classification metadata travels with state coherently across distribution.
- **A — Agentically Engineered.** Classification enforcement is agentic within governance-defined policy.
- **E — Executed.** Continuous classification verification at every state-movement event.
- The exponent — the closure — is the assign-propagate-review loop.

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[10] Andrei Sabelfeld and Andrew C. Myers, "Language-Based Information-Flow Security," *IEEE Journal on Selected Areas in Communications* 21, no. 1 (2003): 5–19, https://doi.org/10.1109/JSAC.2002.806121.

[11] Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

The remaining EDO papers (accreditation, interoperability, deployment) operate on the classification substrate this paper specifies.[12][13]

References

— end v1.0-preprint-draft —

Drafting state: v1.0-preprint draft. PR count: 12 inline (Bell-LaPadula MITRE + McLean SP · Biba MITRE · Hutchinson C&S · Solove CLSR · Liu TDSC · Sabelfeld JSAC · Newsome NDSS · Aleti ACM CSUR · Gelenbe FGCS · Rubin IEEE Computer · NIST + supporting). Existing draft `drafts/edo-wave-2/P8-classification-problem.md` folds into v0.5 expansion.

Appendix — v1.1 Errata Addendum

Added 2026-05-17 per Round-2 EDITORIAL-STAMP HOLD-substance verdict. The v1.0-preprint remains canonical at its original DOI; this v1.1 supplements with the corrections below.

§X. v1.1 Errata — Round-2 Substance Corrections

§X.1. Why this addendum exists

EDO-P8 v1.0-preprint shipped on 2026-05-01 at DOI 10.5281/zenodo.20180168 and was published reader-visible on nonsequitur.tech/pubs/white-papers/classification-problem/. The paper specifies distributed-state classification integrity for agentic operation on the tactical substrate. On 2026-05-17, the post-publish Round-2 editorial review (stamper: Edna; verdict: HOLD-substance; parent thread: YKS-Spine-Binder#355) identified substance defects in the published text that require correction.

The Round-2 review found: (1) §4 (Classification as Continuous Process) contains a substance deficit — the assign-propagate-review loop is named but not animated against the substrate; (2) §5 (Skipjack Coupling) contains a substance deficit — the four coupling mechanisms are listed without operational development, rendering the section redundant with §3; (3) §6 (HGC^3AE^2 return) omits the second E (E^2), walking only seven of eight positional letters; (4) the citation apparatus contains five orphaned references never cited in the body text, one marginal citation used to support an engineering claim, and an inflated peer-reviewed count in the drafting-state footer.

This addendum does not revoke v1.0. It ships as a companion artifact with its own Zenodo record. Readers encountering v1.0 should read this addendum for the canonical corrections. The v1.0 DOI remains stable; the v1.1 record references it as the corrected artifact.

[12] EDO-P9 *Accreditation Problem* promotion (in concurrent drafting; yks-build#108).
[13] EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).

§X.2. Defect: §4 substance deficit — assign-propagate-review loop not animated

v1.0 said: "Three phases. Assignment at state creation… Propagation at every state movement… Re-evaluation at sprint cadence: classification is reviewed against current operating conditions; changes are propagated through the authority chain." (§4, complete text — three sentences.)

The problem: §6 claims "the exponent — the closure — is the assign-propagate-review loop." A paper whose conclusion load-bears on this loop must develop the loop in §4 with operational specificity. The Round-2 review identified three unaddressed questions: (a) what triggers re-evaluation beyond sprint cadence (e.g., substrate partition, authority-chain conflict, external intelligence); (b) what happens when re-evaluation changes a classification level while propagation of the prior level is still in-flight across DTN bundles (EDO-P3) [FORWARD — pending v1.0]; (c) how the assign-propagate-review loop resolves conflicting classification decisions from multiple authority chains operating in a partitioned network where the authority chain specified in §3 cannot be consulted synchronously.

Canonical correction: §4 must be expanded to address, at minimum:

1. **Re-evaluation triggers.** Re-evaluation fires on three conditions: (a) sprint-cadence scheduled review, (b) substrate-boundary event that surfaces a classification mismatch (detected per §3 spillage-detection mechanism), and (c) authority-chain directive propagated through the governance layer. Each trigger produces a re-evaluation record signed by the re-evaluating authority with the same attestation discipline as the original assignment (§3, classification authority chain).
2. **Mid-propagation reclassification.** When a classification level changes while prior-level state is in transit across DTN bundles, the reclassification event produces a revocation bundle that propagates through the same routing substrate. Receiving nodes that have already accepted the prior-level state treat the revocation as a spillage-equivalent event: the state is quarantined pending re-classification verification. The latency window between reclassification and revocation-receipt is an acknowledged exposure; its bound is the DTN propagation delay specified in EDO-P3 [FORWARD — pending v1.0].
3. **Partition-conflict resolution.** In a partitioned network, multiple authority chains may independently reclassify the same state. On partition heal, the conflict is resolved by authority-chain precedence: the highest-authority reclassification wins. If authorities are equal, the more restrictive classification (higher level) wins. This is the conservative-default principle: ambiguity resolves to restriction, not permissiveness. The merge record is auditable and persists in the classification authority chain.

Rationale: the assign-propagate-review loop is the paper's claimed closure mechanism (§6, exponent). Closure mechanisms that are named but not animated are shape without substance — precisely the defect class Round-2 exists to catch.

§X.3. Defect: §5 substance deficit — Skipjack coupling is redundant shape

v1.0 said: "Context integrity is enforced through classification verification at substrate boundaries. Structured routing uses classification as a routing-decision input… HITL control points fire on spillage detection. Iterative feedback closes via classification review at sprint ceremony." (§5, complete text — four sentences, one per Skipjack mechanism.)

The problem: Each sentence restates a §3 architectural commitment through a Skipjack lens without adding mechanism. The Round-2 review asked: "What does Skipjack's context-integrity mechanism actually do with classification metadata that differs from what §3 already specified?" The answer in v1.0 is: nothing — §5 is a relabeling of §3, not a coupling specification.

Canonical correction: The author must choose one of two remediation paths:

Path A — Develop the coupling. Expand at least two of the four Skipjack mechanisms with operational specificity that goes beyond §3. For example: (a) **Structured routing with classification as constraint:** Skipjack's routing algorithm (MR-P5, zenodo.org/records/19869293) accepts classification level as a hard routing constraint — a bundle classified at level L is routable only to substrates whose classification clearance ≥ L. This is NOT the same as §3's "bundles cannot be propagated to substrates whose classification level does not admit them" — it specifies WHERE in the Skipjack routing decision the constraint binds (route selection, not post-hoc rejection). (b) **HITL escalation protocol on spillage:** when §3's spillage-detection mechanism fires, Skipjack's HITL control point does not merely "fire" — it halts propagation on the affected routing path, surfaces the spillage event to the human governance layer with a quarantine recommendation, and blocks path resumption until the human operator issues a signed clearance or re-classification directive.

Path B — Collapse into §3. Acknowledge that the classification integrity architecture (§3) IS the Skip